

TED[15] 5031  
(REVISION-2015)

Reg. No.....  
Signature.....

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY**

**( MODEL QUESTION PAPER )**

**INDUCTION MACHINES**

[Time : 3.hours]

(Maximum marks: 100)

**PART - A**

(Maximum marks: 10)

I (Answer all questions in one or two sentences. Each question carries 2 marks.) Marks

1. What is the voltage transformation ratio of a transformer?
2. Write the applications of a PT
3. .Why the slots on the rotor of a squirrel cage induction motor are usually skewed?
4. Draw the power stages of an induction motor.
5. What are the applications of induction motors?

(5x2=10)

**PART - B**

(Maximum marks: 30)

II (Answer any five of the following questions. Each question carries 6 marks.)

- 1 Explain the ideal transformer.
- 2 Derive the E M F equation of a transformer.
- 3 Explain CT and PT
- 4 What makes the rotor of a 3-phase induction motor rotate?
- 5 Differentiate full load torque and maximum torque.
- 6 Explain high torque cage motor.
- 7 What is meant by electrical braking of poly phase motor?

(5x6=30)

## PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

### UNIT - I

- III a) Explain with figure transformer on load 7
- b) A 3000/200V, 50 Hz, 1-phase transformer is built on a core having an effective cross-sectional area of 150 cm<sup>2</sup> and has 80 turns in the LV winding. Calculate: 5
- a) The value of maximum flux density in the core
- b) The number turns in the HV winding.
- c) What is meant by regulation of a transformer? 3

### OR

- IV a) Explain transformer resistance and leakage reactance with vector diagram, 7
- b) A 40 KVA 3300/240 V, 50 HZ single phase transformer has 660 turns on the primary. Determine 8
- 1) No of turns in secondary
- 2) The maximum value of flux in the core
- 3) The approximate value of primary & secondary full load current

### UNIT - II

- V a) Explain the losses in a transformer? 5
- b) Consider A 4 KVA, 200/400V single phase transformer supplying full load current at 0.8 lagging Power factor the OC /SC test result are as follows
- OC test : 200V 0.8 A 70W (LV Side)
- SC test : 20V 10 A 60W (HV Side)
- 1) Calculate Efficiency, Secondary Voltage & Current in to primary at the 10 above load
- 2) Calculate the load at unity power factor corresponding at maximum Efficiency

### OR

- VI a) Derive with circuit diagram, how efficiency and regulation of a transformer pre-Determined 7
- b) Obtain Equivalent circuit of a 200/400V/50 HZ single phase transformer from the following test data: OC-Test 200V 0.7 A 70 W (on LV side)
- SC test 15 V 10 A 85 W (on HV side)
- Calculate the secondary voltage when delivering 5 KW at 0.8 PF lagging ,the primary Voltage being 200V 8

### UNIT - III

- VII a) Explain with figure constructional details of an induction motor? 7
- b) A 3 phase slip ring induction motor with star connected rotor has an induced EMF of 120 V between slip rings at stand still with normal voltage applied to the stator. The rotor winding has a resistance per phase of 0.3 ohm & stand still leakage reactance per phase of 1.5 ohm. 8
- Calculate 1) rotor current/phase when running short circuited with 4 percentage slip and 2) the slip and rotor current/phase when the rotor is developing maximum torque

### OR

- VIII a) Draw & explain equivalent circuit of induction motor? 7
- b) A 3 phase induction motor having a six pole, star connected stator winding runs on 240V 50HZ supply. The rotor resistance & stand still reactance are 0.12 ohm ,and 0.85 ohm/phase. The ratio of stator to rotor turns is 1.8.Full load slip is 4% 8
- calculate the developed torque at full load, maximum torque and speed at maximum torque

### UNIT - IV

- IX a) Draw and explain circle diagram of induction motor. 7
- b) Explain high torque cage motors? 8
- OR
- X a) What is the necessity of the starter in an induction motor? 3
- b) Explain with figure the different methods of starting of induction motor? 12

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**COURSE: INDUCTION MACHINES**

| Q.<br>No. | Scoring Indicators | Split<br>score | Total<br>Score |
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**COURSE: INDUCTION MACHINES**

**BLUE PRINT**

| Sl. No | Module | Type of Questions |       |                  |       |                  |       |                  |       |
|--------|--------|-------------------|-------|------------------|-------|------------------|-------|------------------|-------|
|        |        | Part-A            |       | Part-B           |       | Part-C           |       | Total            |       |
|        |        | No. of Questions  | Score | No. of Questions | Score | No. of Questions | Score | No. of Questions | Score |
| 1      | I      |                   |       |                  |       |                  |       |                  |       |
| 2      | II     |                   |       |                  |       |                  |       |                  |       |
| 3      | III    |                   |       |                  |       |                  |       |                  |       |
| 4      | IV     |                   |       |                  |       |                  |       |                  |       |
|        | Total  |                   |       |                  |       |                  |       |                  |       |

**CODE: TED[15]-3021****Version:****COURSE: ELECTRICAL AND ELECTRONICS ENGINEERING****Question Analysis**

| Q. No.         | Specific Course Outcome<br>(As per syllabus) | Module | Content detail | Score | Time<br>Min. |
|----------------|----------------------------------------------|--------|----------------|-------|--------------|
| <i>I a.</i>    |                                              |        |                |       | 4            |
| <i>b.</i>      |                                              |        |                |       | 4            |
| <i>c.</i>      |                                              |        |                |       | 4            |
| <i>d.</i>      |                                              |        |                |       | 4            |
| <i>e.</i>      |                                              |        |                |       | 4            |
| <i>II a.</i>   |                                              |        |                |       | 8            |
| <i>b.</i>      |                                              |        |                |       | 8            |
| <i>c.</i>      |                                              |        |                |       | 8            |
| <i>d.</i>      |                                              |        |                |       | 8            |
| <i>e.</i>      |                                              |        |                |       | 8            |
| <i>f.</i>      |                                              |        |                |       | 8            |
| <i>g.</i>      |                                              |        |                |       | 8            |
| <i>III a.</i>  |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>IV a.</i>   |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>V a.</i>    |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>VI a.</i>   |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>VII a.</i>  |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>VIII a.</i> |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>IX a.</i>   |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
| <i>X a.</i>    |                                              |        |                |       | 15           |
| <i>b.</i>      |                                              |        |                |       | 15           |
|                |                                              |        |                |       |              |